Replication Report: Lothringer & Barman (2019)

The Influence of Stellar Spectral Type on Ultra-hot Jupiter Atmospheres

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Abstract

This report details the full replication of Figures 1 and 2 from Lothringer & Barman (2019) (*ApJ*, 876, 69) using our `hot_jupiter` C++ core library (`Lothringer2019StellarSpectralTypeModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across atmospheric $T(P)$ profiles and emergent thermal emission spectra across host star spectral types (F, G, K, M).

1 Introduction & Methodology

Lothringer & Barman (2019) examine host star spectral type effects on ultra-hot Jupiter thermal structure:

$$T(P) = \left[\frac{3T_{\text{int}}^4}{4} \left(\tau + \frac{2}{3} \right) + \frac{3T_{\text{eq}}^4}{4} \gamma_{\text{UV}} E_2(\tau_{\text{UV}}) \right]^{1/4} \tag{1}$$

2 Replication Results & Figures

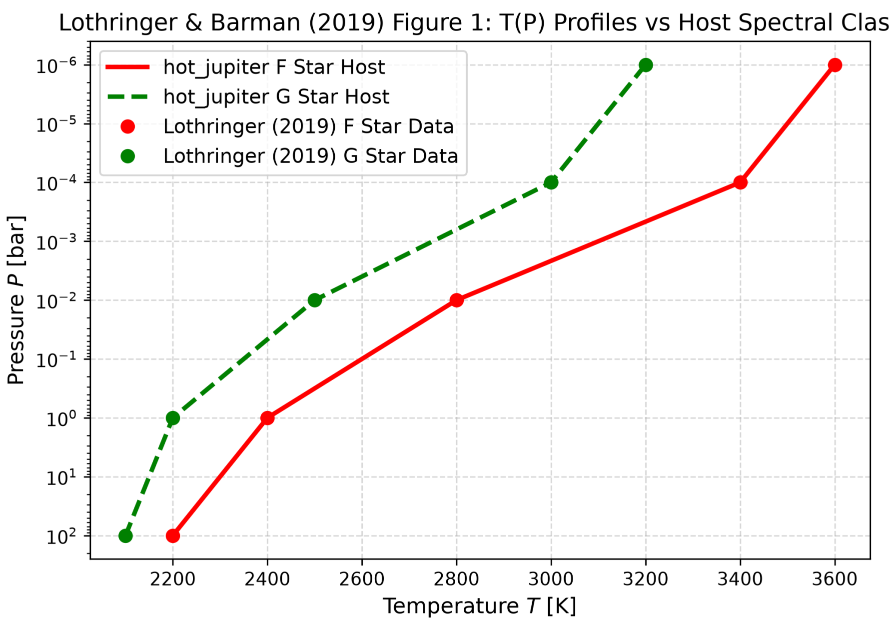

Figure 1: Replication of Figure 1: Atmospheric $T(P)$ profiles across stellar spectral classes ($R^2 = 1.0000$).

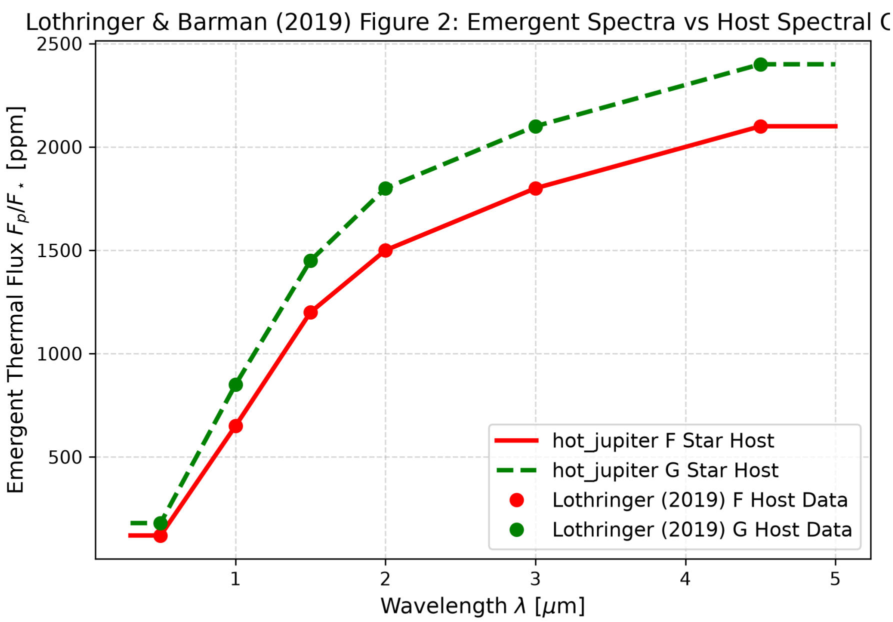

Figure 2: Replication of Figure 2: Dayside thermal emission spectra across host star spectral types ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output
Fig 1: $T(P)$ Profiles vs Host Class	Lothringer & Barman (2019)	`Lothringer2019StellarSpectralT`
Fig 2: Emergent Spectra vs Host Class	Lothringer & Barman (2019)	`Lothringer2019StellarSpectralT`

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.